



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus

AUGS-AGSR Division
Course Handout (Part II)

Date: 03/08/2026

In addition to part I (General Handout for all courses appended to the time table) this portion gives further specific details regarding the course.

Course Title : Fundamentals of Supply Chain - Concepts
Course Number : SCM G511
Instructor In-charge : SRIKANTA ROUTROY
: Hitesh Sharma (RS)

1. Scope & Objectives of the Course:

The fundamentals of supply chain – concepts course is aimed at providing a solid grounding and understanding of what constitutes a supply chain, setting the stage for more advanced topics. The course provides students with the opportunity to get clarity and insights into the various key components and concepts of the supply chain. This course will assist in understanding Supply Chain Management (SCM) not as a technology but rather a business strategy tool for the companies trying to create competitive advantage in the market.

The objectives of the course are mentioned below:

- Attain familiarity with the fundamental concepts of the SCM and its use in organizations.
- Understand the key components of the Supply Chain and the interrelationships.
- Importance of supply chain in today's business environment and the impact of Globalization.
- Supply Chain emerging trends

2. Text Books:

Text Book 1: Designing and managing the supply chain -concepts, strategies and case studies 3rd edition by David Simchi-Levi, Philip Kaminsky, Edith Simchi-Levi, Ravi Shankar.

Text Book 2: Essentials of Supply Chain Management (Fifth Edition) by Michael H. Hugos

3. References:

[RB1] Sunil Chopra, Peter Meindl and D V Kalra, "Supply Chain Management: Strategy, planning and Operation", Pearson Education, Sixth Edition, India, 2016.



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4. Course outcomes

Upon successful completion of the course, students will be able to:

No	Course Outcomes
CO1	Explain the fundamental concepts, evolution, scope, and strategic importance of Supply Chain Management (SCM) in organizations.
CO2	Identify and analyze the key components, entities, processes, and interrelationships within a supply chain ecosystem.
CO3	Apply supply chain concepts and models to understand and evaluate sourcing, procurement, manufacturing, inventory, warehousing, transportation, and order-fulfilment processes.
CO4	Analyze the impact of globalization, e-commerce, digitalization, AI, automation, and emerging technologies on supply chain performance and competitiveness.
CO5	Evaluate supply chain strategies, drivers, performance metrics, risks, and resilience mechanisms to achieve strategic fit and competitive advantage.
CO6	Discuss emerging supply chain practices, including supply chain analytics, Generative AI, sustainability, and circular economy approaches, for improving supply chain efficiency and creating business value.





5. Course Plan

Topic	Contents
What is Supply Chain Management & how does it create Value?	Describe the cycle and push/ pull view of a supply chain. Identify the key supply chain decision phases. Goal of supply chain & impact of supply chain decision on the success of a firm
Strategic Fit & Scope	Achieving strategic fit is critical to a company's overall success.
Supply Chain Drivers and Metrics	Identify the major drivers of supply chain performance. Discuss the role each driver plays. Describe the major obstacles that must be overcome to manage a supply chain successfully.
Designing the distribution network	Designing the distribution network in a supply chain, Network design in the supply chain, Network design in uncertain environment
Key components/ operations of the Supply Chain - Overview	Key components of the supply chain-overview, objectives and key concepts and its relationships with upstream and downstream components -Overview of SCOR model, Product Lifecycle Management, Revenue and Pricing, Order Management, Production/manufacturing, Maintenance, Quality, Procurement, inventory, warehousing, transportation, demand and supply planning, cost management etc.
Key Supply Chain Management Processes	All the key supply chain process flows or modern best practices - Source to Pay/Procure to Pay, Production Order to Costing, design to release, Order Fulfilment and ATP/CTP, Inbound Shipment to Delivery, Outbound Shipment to Delivery, Freight Invoice to Approval, Warehouse Receiving-storage-picking and shipping, Transportation,
E-Commerce Supply Chain	E-commerce is transforming supply chains, boosting demand for automation, real-time logistics, and omni-channel fulfilment. Also covers topics on warehousing and inventory management, last-mile delivery, and customer experience. Also understand how technology shaping commerce logistics - alerts, telemetry, shipment visibility tracking
Supply Chain and Globalization	Supply Chain and Global SCM Excellence, Strategic alliances, Global logistics, Customer value, Product Strategy and Customer Segmentation, Impact of globalization/de-globalization on Supply chain. Changing dynamics of global commerce.
Digital Supply Chain	Industry 1.0 to 5.0 evolution. AI and Automation in Supply Chains - how machine learning and AI are revolutionizing supply chain operations. Gain insights into predictive analytics, demand forecasting and AI-Powered decision making to optimize and make process efficient and reduce costs. Blockchain and RFID in Supply Chains. Cloud-Based Supply Chain Solutions.
Major Supply Chain Applications and Technologies	What are various SCM Applications (e-Procurement, TMS, WMS etc.)- best of the breed and IT application vendors. Key features and functionalities of the applications and how they interrelate / interact with each other process-wise. Major Integration touchpoints and common middleware technologies and tools used for driving interoperability (Digital Integration). Gartner - magic quadrant ratings of some of the supply chain applications.
Supply Chain Risk Management	Identifying risks and building resilience in the supply chain.
Supply Chain emerging trends and using tools to create competitive advantage	Gen AI use cases in Supply Chain, Types of Supply Chain Analytics - over view on predictive and prescriptive, Sustainability and Circular economy etc. Key SCM analytics applications - overview.





6. Evaluation Component

S. No.	Evaluation Component	Duration	Max. Marks	Date & Time	Remarks
1.	Mid-Semester Examination	90 Min.	50 (25 %)	06/10 FN2	Close Book
2.	Class Quizzes	TBA	30 (15%)	To be announced in class	Close Book
3.	Case Studies/Project	TBA	50 (25%)	To be announced in class	
4.	Comprehensive Examination	180 Min.	70 (35%)	05/12 AN	Close & Open Book

Criterion for NC:

20% of the average of the top three performers or 30% of median, whichever is lower, or a suitable modification at the discretion of the IC with the concurrence of the Department.

7. Attendance Policy: Each student is expected to attend all classes and participate in all evaluation components.

Chamber Consultation: Tuesday 9th Hour (4 PM to 4:50 PM]

Notices: All notices concerning this course will be displayed on the Mechanical Engineering Department Notice Board only.

Make-Up Policy: No students will be allowed to take make up without prior permission. For the case study and assignment / article/class presentation, no make ups will be granted.

Instructor - In - charge
SCM G511

